



Rheumatoid Arthritis

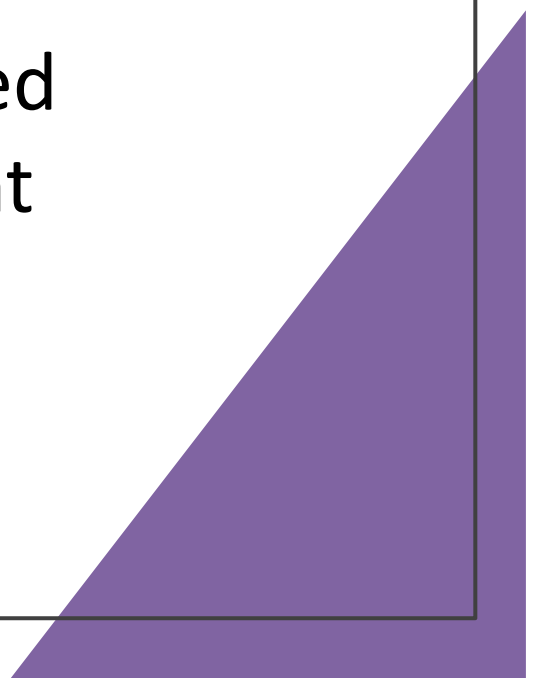
Addis Ababa University

By

Woldu, MA

Definition

- Rheumatoid arthritis (RA) is a chronic, progressive inflammatory disorder of unknown etiology, characterized by polyarticular symmetric joint involvement and systemic manifestations.



Pathophysiology

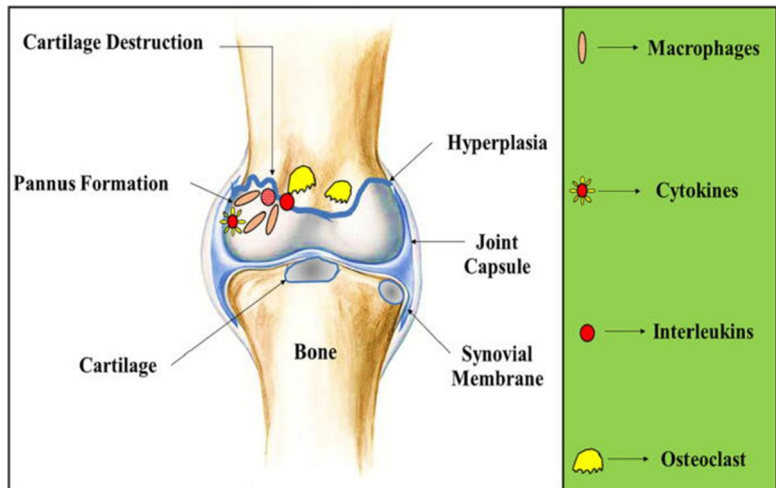
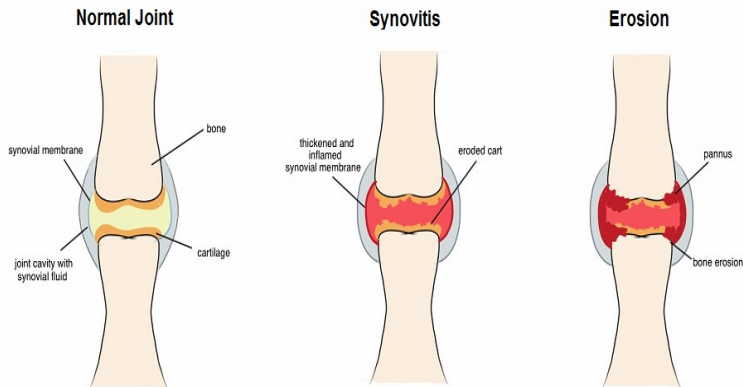
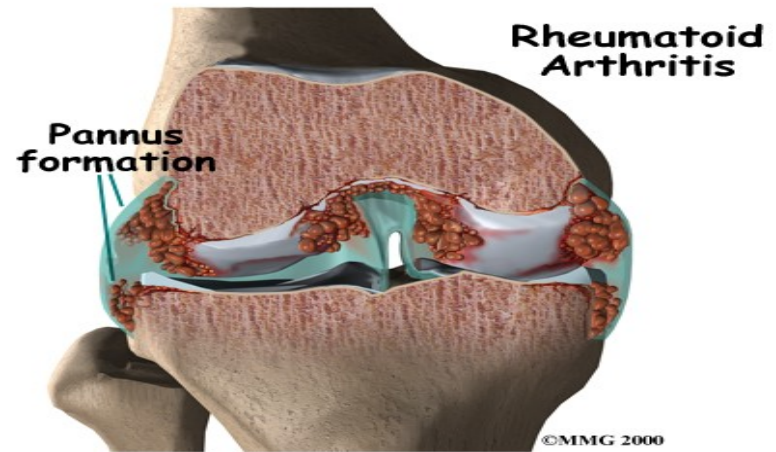
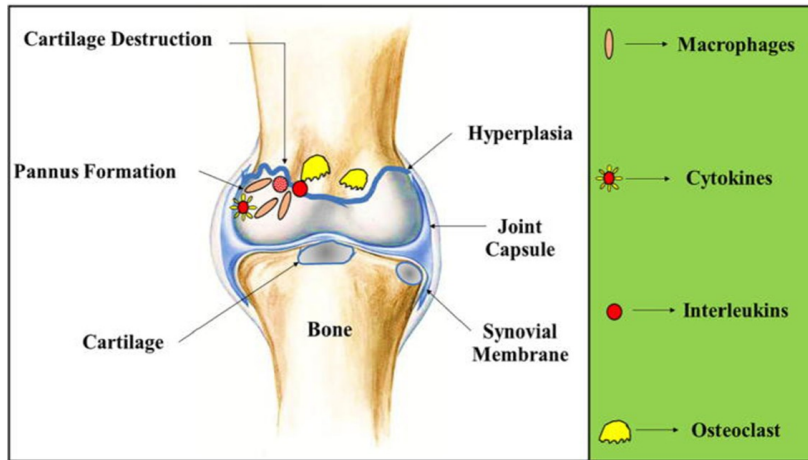
Autoimmune disorder involving dysregulation of T and B cells.

Production of rheumatoid factor and autoantibodies against citrullinated peptides.

Proinflammatory cytokines (TNF- α , IL-1, IL-6) drive inflammation.

Chronic synovial inflammation → pannus formation → cartilage and bone erosion.

Pannus formation



Clinical Presentation

Insidious onset with fatigue, malaise, fever, weight loss.

Symmetric joint involvement (hands, wrists, feet, knees).

Morning stiffness > 1 hour.

Joint swelling, tenderness, warmth.

Chronic deformities: swan-neck, boutonnière, ulnar deviation.

Diagnosis

- **Key Components of RA Diagnosis**
 - **Clinical Symptoms**
 - *Persistent joint swelling*, especially in small joints (hands, wrists, feet)
 - *Morning stiffness* lasting more than 30 minutes
 - *Symmetrical joint involvement*
 - Symptoms lasting *at least six weeks*

Diagnosis...

– **Physical Examination**

- Swelling, tenderness, and reduced range of motion in affected joints
- Rheumatologists often assess multiple joints for inflammation

Diagnosis...

— Laboratory Tests

- **Rheumatoid Factor (RF):** Present in ~70–80% of RA patients, but not exclusive to RA
- **Anti-Cyclic Citrullinated Peptide (anti-CCP):** More specific for RA, often used to confirm diagnosis
- **Erythrocyte Sedimentation Rate (ESR) and C-Reactive Protein (CRP):** Indicate inflammation levels
- Complete blood count to check for anemia or other systemic effects

Diagnosis...

- **Examples of other conditions where RF may be positive (DDX):**
 - Sjögren's syndrome
 - Systemic lupus erythematosus (SLE)
 - Chronic infections (e.g., hepatitis C, tuberculosis)
 - Some healthy elderly individuals
 - So — **RF is a helpful, but not definitive, marker** for rheumatoid arthritis.

Diagnosis...

– Imaging Studies

- **X-rays, ultrasound, or MRI** to detect joint damage, erosion, or inflammation
- Imaging helps differentiate RA from osteoarthritis and other joint disorders

Diagnosis

- American Rheumatism Association (ARA) 1987 Criteria:
 1. Morning stiffness >1 hour.
 2. Arthritis of ≥ 3 joint areas.
 3. Arthritis of hand joints.
 4. Symmetric arthritis.
 5. Rheumatoid nodules.
 6. Serum rheumatoid factor.
 7. Radiographic changes.
- ≥ 4 criteria present for ≥ 6 weeks = RA diagnosis.

Classification Criteria (ACR/EULAR 2010)

- To be classified as RA, patients typically need a score of ≥ 6 out of 10 based on:

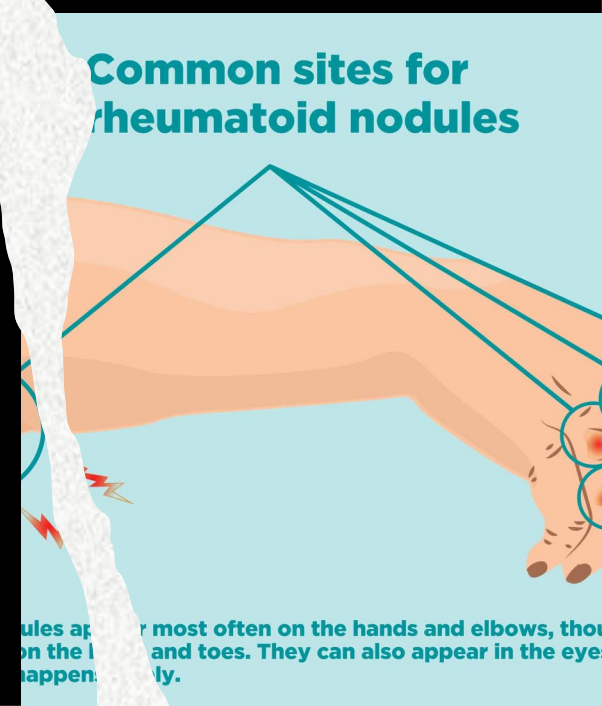
Category	Criteria	Points
Joint involvement	1 large joint = 0; 2–10 large joints = 1; 1–3 small joints = 2; etc.	0–5
Serology (RF, anti-CCP)	Low-positive = 2; High-positive = 3	0–3
Acute-phase reactants	Abnormal CRP or ESR = 1	0–1
Duration of symptoms	≥ 6 weeks = 1	0–1

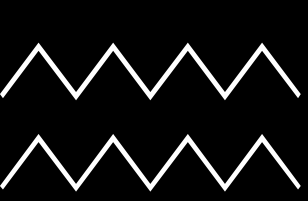
• **ACR:** American College of Rheumatology

• **EULAR:** European Alliance (formerly League) Against Rheumatism



Rheumatoid nodules





Laboratory and Imaging Findings



- Elevated ESR and CRP.



- Positive rheumatoid factor (60–70%).



- Anti-CCP antibodies (anti–cyclic citrullinated peptide)



- X-rays: periarticular osteopenia, joint space narrowing, erosions.

Treatment Goals

- Induce remission or low disease activity.
- Reduce pain, stiffness, and swelling.
- Preserve joint mobility and function.
- Prevent structural damage and disability.
- Improve quality of life.



Non- Pharmacologic Therapy

- Adequate rest and physical therapy.
 - Weight control and joint protection.
 - Occupational therapy and assistive devices.
 - Surgery (synovectomy, tendon repair, joint replacement) in advanced cases.
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Pharmacologic Therapy

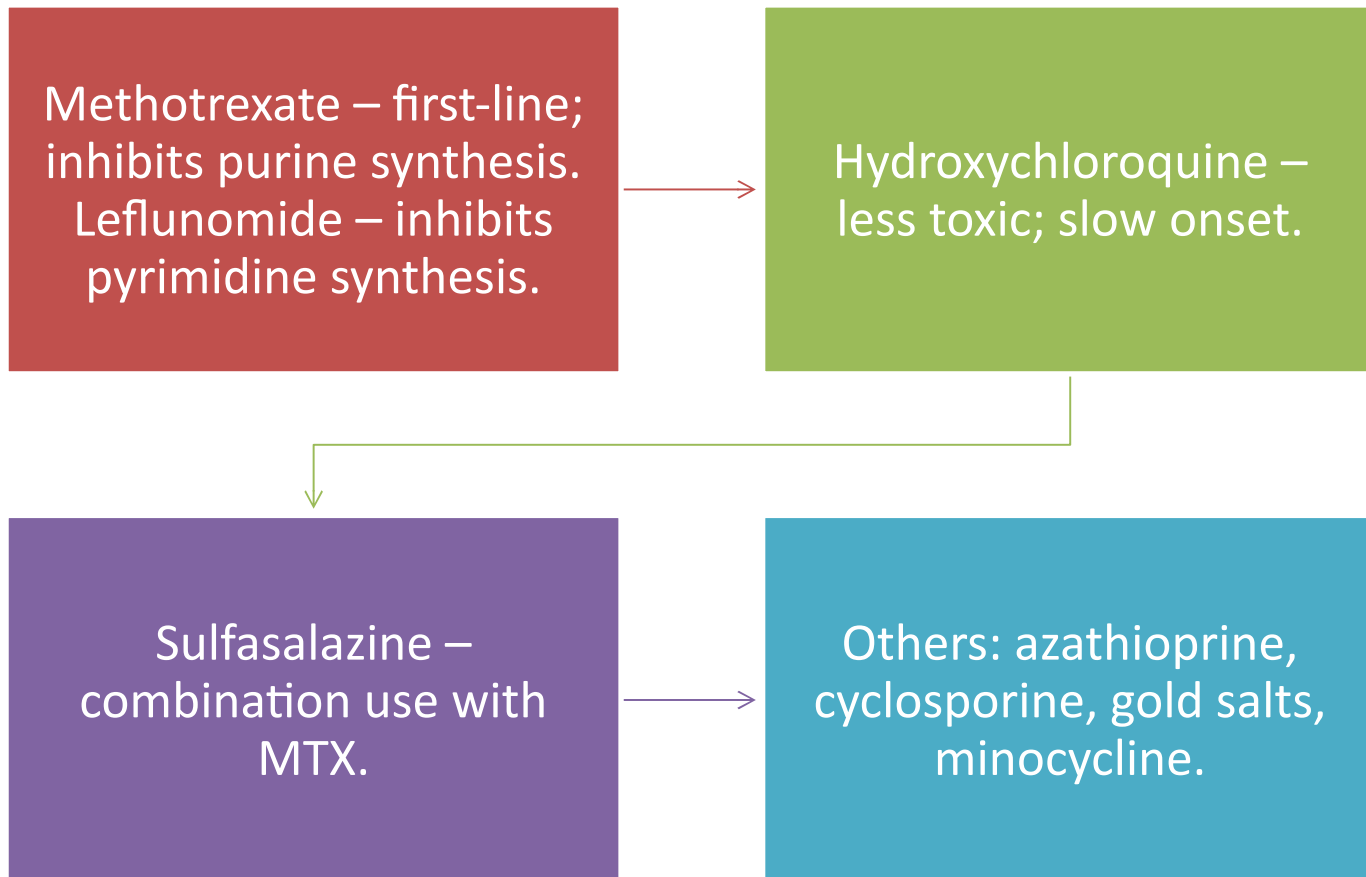
1. NSAIDs: symptomatic relief
(no effect on disease
progression).

2. Corticosteroids: short-term
use for flare control.

3. DMARDs: cornerstone of
RA management.

4. Biologic DMARDs for
inadequate responders.

Disease-Modifying Antirheumatic Drugs (DMARDs)



Methotrexate Dosing

Initial dose: *7.5–15 mg once weekly*, either orally or subcutaneously

Titration: Increased by 2.5–5 mg increments every 2–4 weeks

Maximum dose: *25–30 mg per week* (some guidelines cap at 25 mg)

Route: Oral tablets or subcutaneous injection (injection preferred for better absorption at higher doses)

Methotrexate: Monitoring and Safety

- **Folic acid supplementation:** *1 mg daily* or *5 mg weekly* (on a different day than methotrexate) to reduce side effects
- **Lab monitoring:**
 - CBC, liver enzymes, and renal function every 4–8 weeks
 - Chest X-ray before initiation (to rule out lung disease)
- **Common side effects:** Nausea, fatigue, mouth ulcers, elevated liver enzymes
- **Serious risks:** Hepatotoxicity, bone marrow suppression, interstitial lung disease

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- Why Folic acid and methotrexate not administered together?

Leflunomide

- Leflunomide is an immunomodulatory medication primarily used as a **DMARD** and may take **4–6 weeks** to show clinical effect.
 - It works by **inhibiting pyrimidine synthesis**, which reduces the proliferation of activated **T and B lymphocytes**, thereby decreasing inflammation and joint damage
 - Can be used **alone or in combination** with methotrexate (with careful monitoring for hepatotoxicity).

Leflunomide dosing

- **Loading Dose (Optional)**
 - **100 mg orally once daily for 3 days**
 - Purpose: Rapidly achieve therapeutic plasma levels
 - Note: Some clinicians may **omit the loading dose** to reduce the risk of side effects, particularly gastrointestinal symptoms.

Leflunomide dosing...

- **2. Maintenance Dose**
 - **20 mg orally once daily** (standard maintenance dose)
 - **Alternative: 10 mg once daily** may be used in patients who do not tolerate 20 mg, especially in elderly or those with hepatic concerns.

Leflunomide monitoring

- **Monitoring**
 - **Liver function tests (LFTs):** baseline, then every 2–4 weeks for the first 6 months, then periodically
 - **Complete blood count (CBC):** to monitor for cytopenias
 - **Pregnancy:** Leflunomide is **teratogenic**; women of childbearing age must use effective contraception, and cholestyramine washout is required if pregnancy is planned or occurs.

Hydroxychloroquine

- Hydroxychloroquine is generally **well-tolerated** and safe for long-term use.
 - Not recommended in **patients with pre-existing retinal disease** or **severe renal impairment** without careful monitoring.
 - Safe in **pregnancy** and **lactation** with usual doses, making it preferable in women of childbearing age.
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Hydroxychloroquine (Plaquenil®)

Dosing

- **Standard Adult Dose**
 - **200–400 mg orally once daily**
 - Can also be given as **200 mg twice daily** depending on patient tolerance and convenience.
 - Maximum **daily dose: 5 mg/kg/day** based on **actual body weight** to reduce the risk of **retinopathy**.
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Hydroxychloroquine (Plaquenil®): **Onset of Action**

- **Onset of Action**
 - Effects are **gradual**; clinical improvement typically appears after **6–12 weeks**.
 - Often used in combination with other **DMARDs** such as methotrexate or sulfasalazine.
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Hydroxychloroquine (Plaquenil®): Monitoring

Ophthalmologic exam: baseline, then annually after 5 years of therapy (earlier if risk factors exist)

Blood counts and liver function tests: periodically

Monitor for skin reactions, GI upset, and neuromuscular symptoms

Sulfasalazine (Azulfidine®)

- **1. Indications**

- Disease-modifying antirheumatic drug (**DMARD**) for **mild to moderate RA**
- Often used in **combination therapy** with methotrexate, hydroxychloroquine, or leflunomide (triple therapy)

Sulfasalazine (Azulfidine®): Dosing

2. Dosing

Adults:

- **Starting dose:** 500 mg orally once or twice daily
- **Titration:** Increase by 500 mg every 1–2 weeks as tolerated
- **Usual maintenance dose:** 2–3 grams per day, divided into 2–3 doses
- **Maximum dose:** 3 grams/day

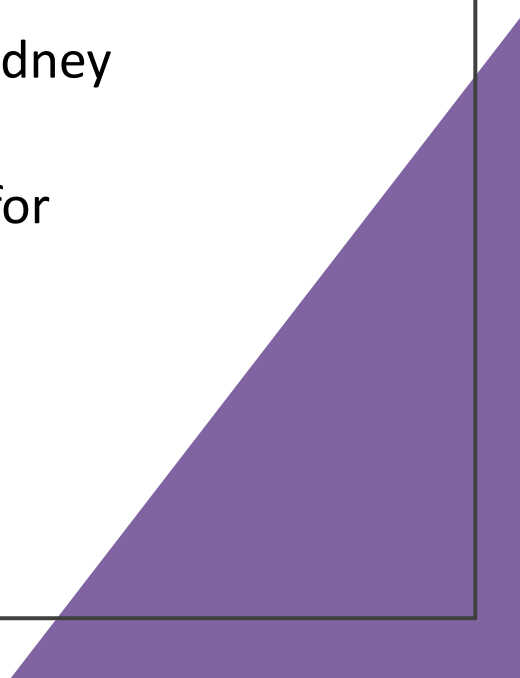
Notes:

- Enteric-coated formulations are available to **reduce gastrointestinal side effects.**
- Take with **food or plenty of fluids** to improve tolerance.

Sulfasalazine (Azulfidine®): Onset of Action

- **Onset of Action**
 - Clinical improvement usually occurs **4–6 weeks**, with maximal effect in **3–6 months**.

Sulfasalazine (Azulfidine®): Monitoring

- **Monitoring**
 - **Baseline:** CBC, liver function tests (LFTs), kidney function
 - **Ongoing:** CBC and LFTs every 2–3 months for the first 3–6 months, then periodically
 - Monitor for **GI upset, rash, headache, and reversible oligospermia** in men
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Sulfasalazine
(Azulfidine®):
Adverse
Effects

- **5. Adverse Effects**
 - Gastrointestinal: nausea, vomiting, anorexia
 - Rash or hypersensitivity reactions
 - Hematologic: leukopenia, anemia
 - Hepatic toxicity (rare)
 - Reversible **male infertility** (usually resolves after discontinuation)

Biologic DMARDs

- TNF inhibitors: etanercept, infliximab, adalimumab.
- IL-1 receptor antagonist: anakinra.
- B-cell depletion: rituximab.
- T-cell co-stimulation modulator: abatacept.
- IL-6 inhibitor: tocilizumab.

Treatment Algorithm

- Stepwise approach:
 - Start DMARD (e.g., methotrexate) ± NSAID ± prednisone within 3 months.
 - If poor response → switch DMARD or combine (MTX + sulfasalazine + hydroxychloroquine).
 - If inadequate response → add biologic DMARD.
 - Persistent disease → consider second-line or combination biologic therapy.

Monitoring

- CBC, liver function tests, and creatinine for MTX and leflunomide.
- Ophthalmologic exams for hydroxychloroquine.
- Screen for TB before biologic therapy.
- Monitor infection signs during biologic/DMARD therapy.

References

- Pharmacotherapy Handbook, 7th Edition – Chapter 4: Rheumatoid Arthritis.
- American College of Rheumatology (ACR) Guidelines.
- Recent reviews on RA pharmacologic management.

